Windows Server Administration
Active Directory, DNS & Group Policy Documentation

Hands-on lab project simulating enterprise-grade Windows Server environment

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Domain: dhvanit.local.com | Platform: Windows Server 2019 | Tool: VirtualBox
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1. Project Overview

This project documents the setup and configuration of a Windows Server 2019 environment in an isolated VirtualBox home lab. The goal was to simulate a real enterprise network with a Domain Controller, Active Directory, DNS, and Group Policy enforcement.

Component	Purpose
Active Directory Domain Services (AD DS)	Centralized user and department management
DNS Server	Internal domain name resolution
Group Policy Objects (GPO)	Enforce security policies across domain
Organizational Units (OUs)	Separate HR and Admin departments logically
User Account Management	Create and manage domain user accounts

2. Lab Environment

2.1 Virtual Machine Setup

VM	OS	Role	IP Address
Server VM	Windows Server 2019	Domain Controller	192.168.56.10
Client VM 1	Windows 10 Pro	Domain Member	192.168.56.20
Client VM 2	Windows 10 Pro	Domain Member	192.168.56.21
Attacker VM	Kali Linux	Security Testing	192.168.56.30

2.2 Domain Information

- Domain Name: dhvanit.local.com
- Domain Controller Hostname: DC01
- Network: Host-Only (192.168.56.0/24) — fully isolated
- DNS Server: Configured on Domain Controller (192.168.56.10)

■ *All machines isolated in a private virtual network.*

3. Active Directory Configuration

3.1 Domain Setup

```
Server Manager → Add Roles and Features → Active Directory Domain Services
→ Promote to Domain Controller → Add new forest: dhvanit.local.com
→ Set DSRM password → Install → Restart
```

3.2 Organizational Units & Users

OU	Users Created	Role	Access Level
OU=HR_Department	hr.user1, hr.user2	HR Staff	Limited — no system changes
OU=Admin_Department	admin.user1, admin.user2	IT Admin	Elevated — system management
OU=Workstations	All machines	Computers	Machine-level GPO applied
OU=Disabled_Users	Inactive accounts	Deactivated	No login access

4. Group Policy Objects (GPO)

GPO Name	Applied To	Key Settings
Password Policy GPO	Default Domain	Min 8 chars, complexity required, lockout after 5 attempts, 30-day expiry
Firewall Policy GPO	OU=Workstations	Firewall ON for all profiles, inbound RDP blocked
Recycle Bin GPO	OU=HR_Department	Recycle Bin shortcut on Desktop via Group Policy Preferences

```
C:\> gpupdate /force
C:\> gpresult /r
```

■ *All GPOs verified by logging into client VMs and running gpresult /r*

5. Errors Encountered & Resolved

Error 1: DNS not resolving domain name

Cause: DNS server not set to DC IP on clients

Resolution: Manually set Primary DNS to 192.168.56.10 on all client VMs

Error 2: GPO not applying on client machine

Cause: Group Policy not refreshed after change

Resolution: Ran gpupdate /force on client — policies applied successfully

Error 3: User unable to join domain

Cause: Firewall blocking domain traffic

Resolution: Temporarily disabled firewall, joined domain, then re-enabled GPO firewall

Error 4: DSRM password prompt during DC promo

Cause: Incorrect password entered during setup

Resolution: Re-ran dcpromo with correct DSRM password and reinstalled AD DS role

Error 5: Recycle Bin GPO not visible in GPMC

Cause: Confusion between GPO Policies vs Preferences

Resolution: Used Group Policy Preferences under User Config to deploy Recycle Bin shortcut

■ *Error 3 insight: Always join domain before enforcing strict firewall GPOs.*

6. Security Observations

#	Observation	Risk	Action Taken
1	Default Administrator account active	HIGH	Renamed to non-obvious name
2	No password complexity by default	HIGH	Enforced via Password Policy GPO
3	No account lockout policy	HIGH	Set lockout after 5 failed attempts
4	RDP port 3389 open internally	MEDIUM	Blocked inbound RDP via Firewall GPO
5	LDAP unencrypted on port 389	MEDIUM	Noted for future LDAPS upgrade

7. Skills Gained

- Installing and promoting Windows Server 2019 as a Domain Controller
- Configuring Active Directory with custom domain dhvanit.local.com
- Creating OUs, Users, and Security Groups for HR and Admin departments
- Implementing Role-Based Access Control (RBAC) through security groups
- Configuring GPOs — Password Policy, Firewall, and Recycle Bin
- Troubleshooting real domain errors — DNS, GPO, domain join, DSRM
- Identifying and remediating Active Directory security misconfigurations

8. Disclaimer

All configurations were performed in a privately owned, isolated VirtualBox lab. No real systems were accessed. Created for educational purposes only.
